组会报告

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Motivation

Li, Ying & Zhao, Hui & Zhang, & Hu,. (2019). Detection of Wildfire Smoke Images Based on a Densely Dilated Convolutional Network. Electronics. 8. 1131. 10.3390/electronics8101131.

烟雾检测是识别潜在火灾的有效方法。

烟雾检测有着不小的进展，背景处理展示了 CNN 可以处理复杂背景。但是不能直接应用于野火烟雾检测。

存在的问题：1.像烟的区域类似雾，云，霾，因此难以区分。2.缺乏野火烟雾图像。

解决方法：提出集成传统方法的 CNN，包含一个候选烟雾区域分割策略和空洞 DenseNet。RGB 图像转换为 YUV 色彩空间，确保颜色信息不受光照变化影响。

Method

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候选烟雾区域分割：大量真实烟雾图像的概率统计表示，非烟区域的 U-V 值主要分布在 0 - 40，烟雾区域主要分布在 40 - 130。

$$S_1(x,y)=\begin{cases}1 & if\ |U(x,y)-128|>T_U \\ 0 & Otherwise\end{cases}$$

$$S_2(x,y)=\begin{cases}1 & if\ |U(x,y)-V(x,y)|>T_{UV} \\ 0 & Otherwise\end{cases},$$

T 为阈值，由多次试验得到

最终候选区域为

$$S_0(x,y)=\begin{cases}I_i(x,y) & if\ S_1(x,y)=1(or)S_2(x,y)=1 \\ 0 & Otherwise\end{cases},$$

Method

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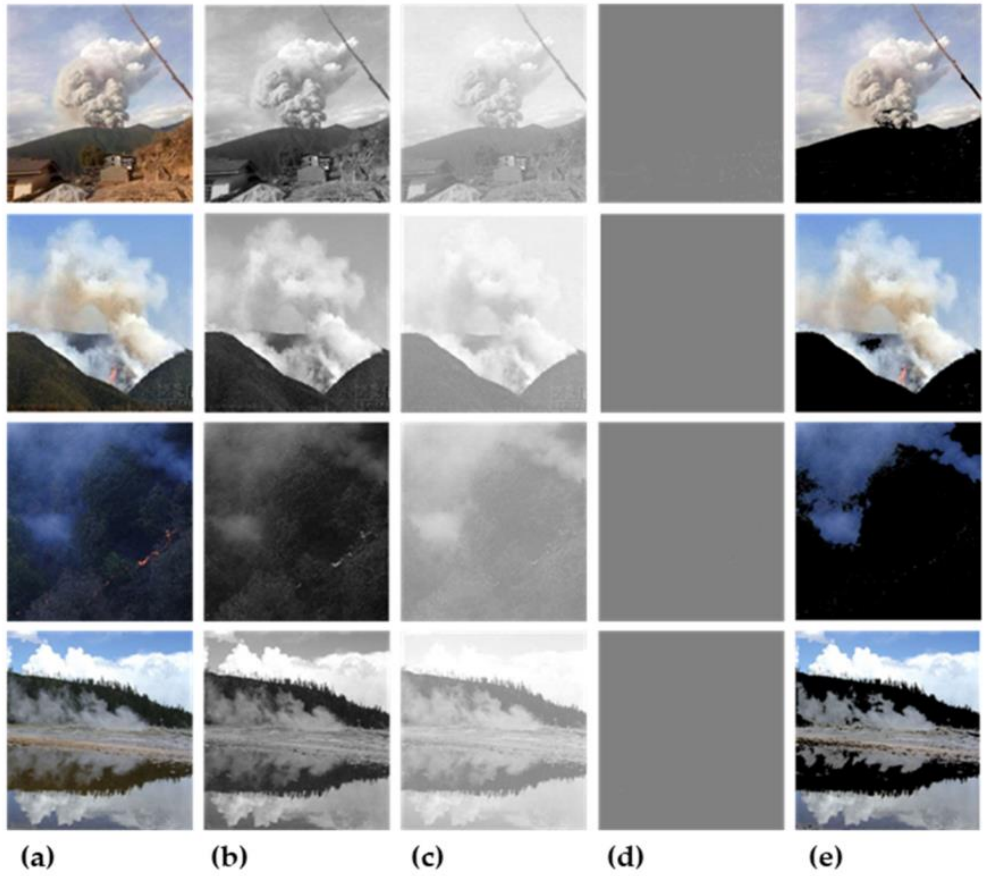

Figure 1. Segmentation results in the YUV color space. (**a**) RGB image, (**b**) Y component, (**c**) U component, (**d**) V component, and (**e**) candidate smoke region.

烟雾候选区域分割结果

Architecture

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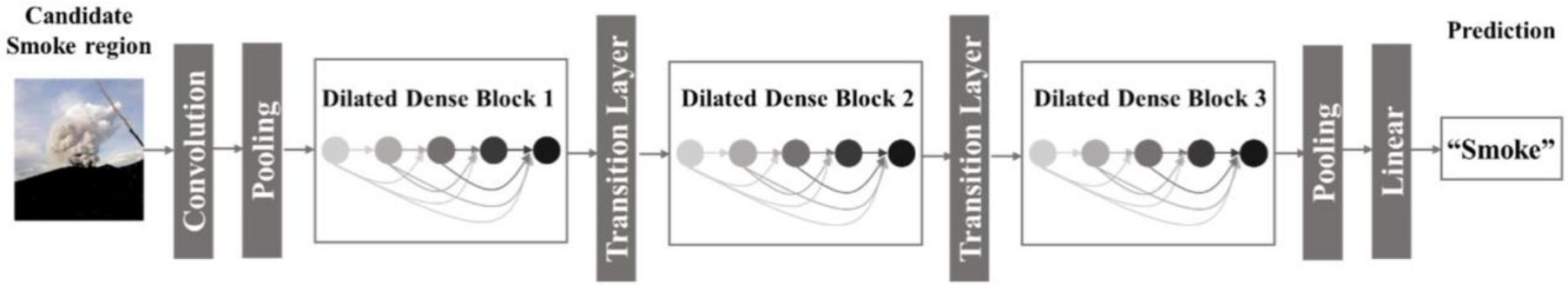

Figure 2. Network architecture of proposed wildfire smoke dilated DenseNet (WSDD-Net).

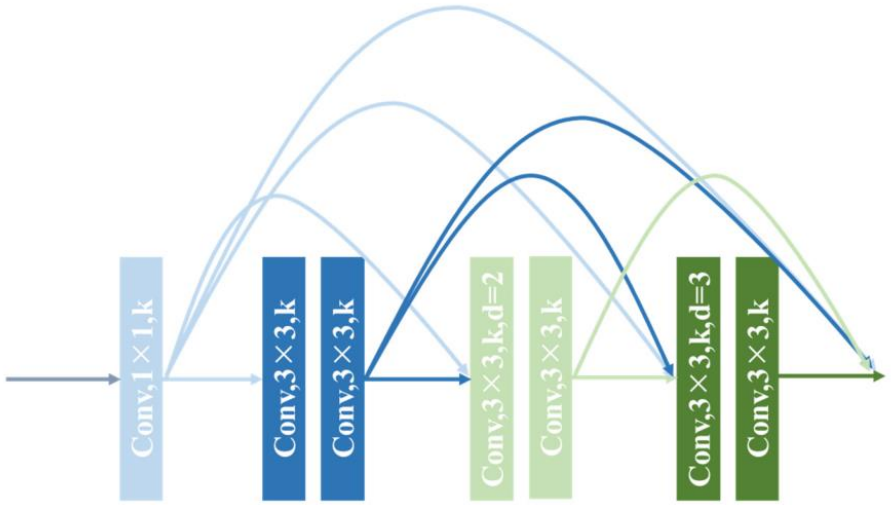

Figure 4. Framework of the proposed dilated dense connection (DDC).

Dilated Dense Block

$$BCE(p,y)=\begin{cases}-\alpha\log p & y=1\\ -(1-\alpha)\log(1-p) & y=0\end{cases}.$$

解决类别不平衡，使用平衡交叉熵

Experiment

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Table 1. Comparisons with classical deep convolutional neural networks (CNNs).

Model	Size (MB)	AR (%)	FAR (%)	F1	RR (%)	DR (%)	Test time (ms)
ResNet-50[30]	89.69	96.02	2.41	96.23	97.85	94.66	10.31
ResNet-101[30]	162.14	95.54	7.56	95.94	93.77	98.22	12.89
Inception-V3[31]	83.12	91.72	15.12	92.68	88.20	97.63	10.70
DenseNet-121[22]	26.53	96.34	4.47	96.60	96.18	97.03	8.17
DenseNet-169[22]	47.64	97.29	2.75	97.47	97.62	97.33	10.07
WSDD-Net	39.72	99.20	0.34	99.25	99.70	98.81	9.93